AI Personal Operations Manager

GOOGLE CLOUD GEN AI ACADEMY APAC 2026

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Challenge: Build a multi-agent AI system that helps users manage tasks, schedules, and information by interacting with multiple tools and data sources.

THE BIG IDEA

From Intent to Execution — Automatically

The **AI Personal Operations Manager** is an intelligent multi-agent system designed to act as a unified productivity assistant. It understands natural language and autonomously manages tasks, schedules, notes, and reminders — all from a single interface.

Understands You

Interprets natural language goals and converts them into structured, actionable workflows.

Coordinates Agents

Deploys specialized AI agents in concert to handle planning, scheduling, notes, and reminders.

Executes & Tracks

Runs tools, stores structured outputs in BigQuery, and delivers a full workflow trace.

SOLUTION EXPLANATION

A Coordinated Multi-Agent Architecture

The Agent Hierarchy

A central **Orchestrator Agent** classifies user intent and delegates to specialized sub-agents, each owning a distinct domain:

- **Planner**
  Breaks high-level goals into granular tasks
- **Calendar**
  Schedules events with intelligent time suggestions
- **Notes**
  Summarizes content and extracts action items
- **Reminder**
  Evaluates urgency and triggers alerts

Core Workflow

Example: *"Plan my exam week"* → Tasks generated → Automatically scheduled → Stored in BigQuery → Returned as a structured plan with full trace.

OPPORTUNITIES & USP

Beyond Chatbots — This System Acts

Most AI assistants stop at suggestions. The AI Personal Operations Manager goes further — it executes, coordinates, and delivers real-world outcomes through a fully integrated agent network.

True Multi-Agent Orchestration

Agents collaborate dynamically, not in isolation.

End-to-End Automation

From intent capture to data storage — zero manual steps.

MCP-Style Tool Integration

Clean abstraction layer connects agents to real tools.

Cloud-Native & Scalable

Built on GCP for elastic, production-grade performance.

Workflow Trace Visualization

Real-time visibility into every agent decision and action.

FEATURES

A Full Suite of Intelligent Capabilities

Natural Language Interaction

Users communicate in plain English. The system interprets intent and acts — no commands or templates required.

Intelligent Task Decomposition

Complex goals are automatically broken into prioritized, trackable sub-tasks with deadlines and ownership.

Automated Scheduling

The Calendar Agent suggests optimal time slots and creates events — fully automated and context-aware.

Notes Summarization & Action Extraction

Meeting notes and text input are distilled into summaries with clear, extracted action items ready for follow-up.

Reminder & Urgency Detection

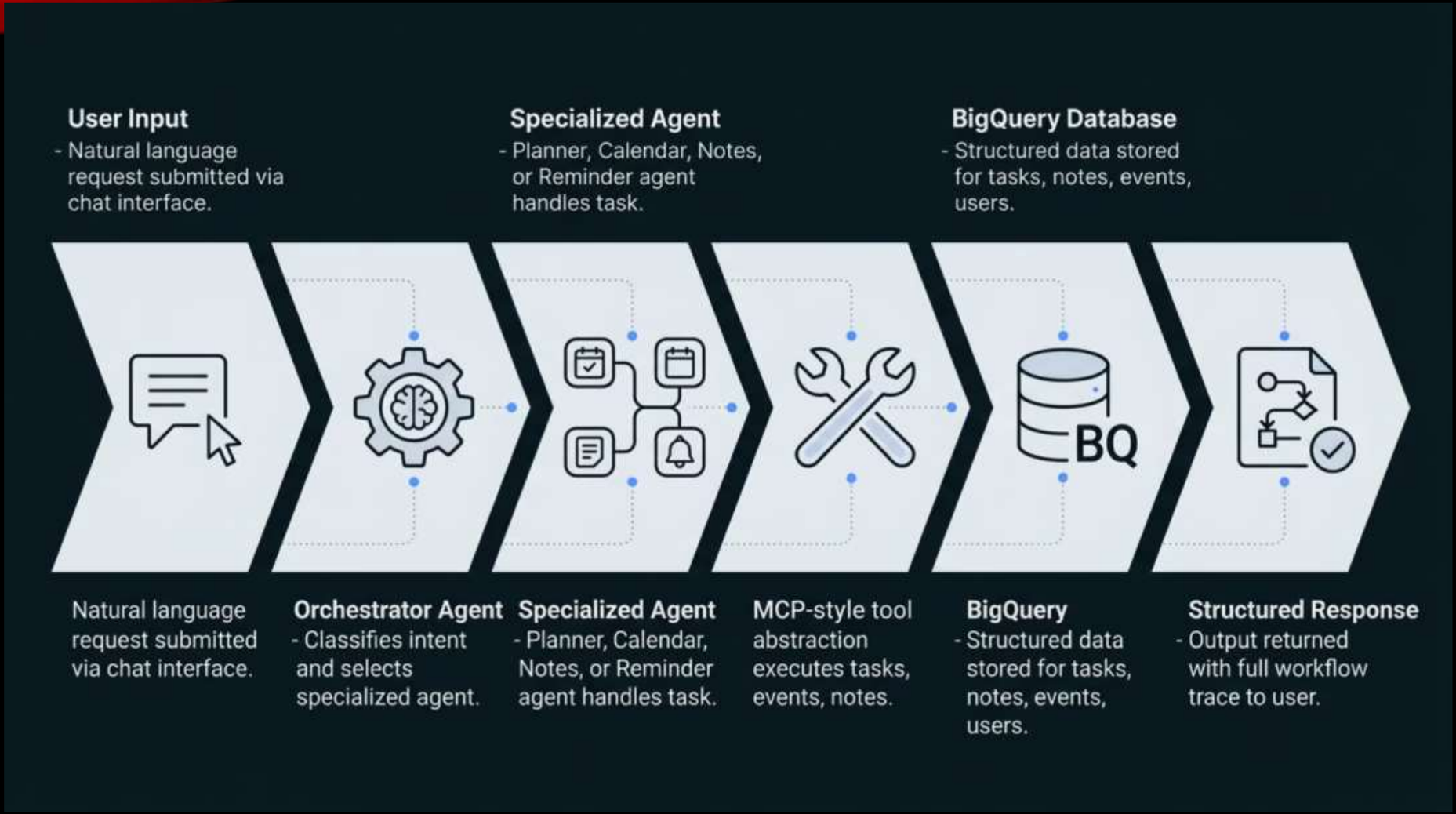
The Reminder Agent proactively evaluates deadlines and surfaces high-priority items before they become blockers.

Resilient Fallback Mode

The system remains fully operational even without GCP connectivity, ensuring uninterrupted productivity.

PROCESS FLOW

How It Works: End-to-End Process



Example use case: A user types **"Plan my exam week"** → The Orchestrator routes to the Planner Agent → Tasks are generated and handed to the Calendar Agent → Events are created and stored in BigQuery → The user receives a complete structured study plan with a full agent trace showing every step taken.

UI / UX

Designed for Clarity and Real-Time Interaction

Key UI Components

💬 Chat Interface

The primary interaction layer. Users submit natural language input and receive structured, actionable responses in real time.

📊 Dashboard

A unified view of tasks, notes summaries, upcoming events, and live system health status for at-a-glance productivity management.

🔀 Workflow Visualizer

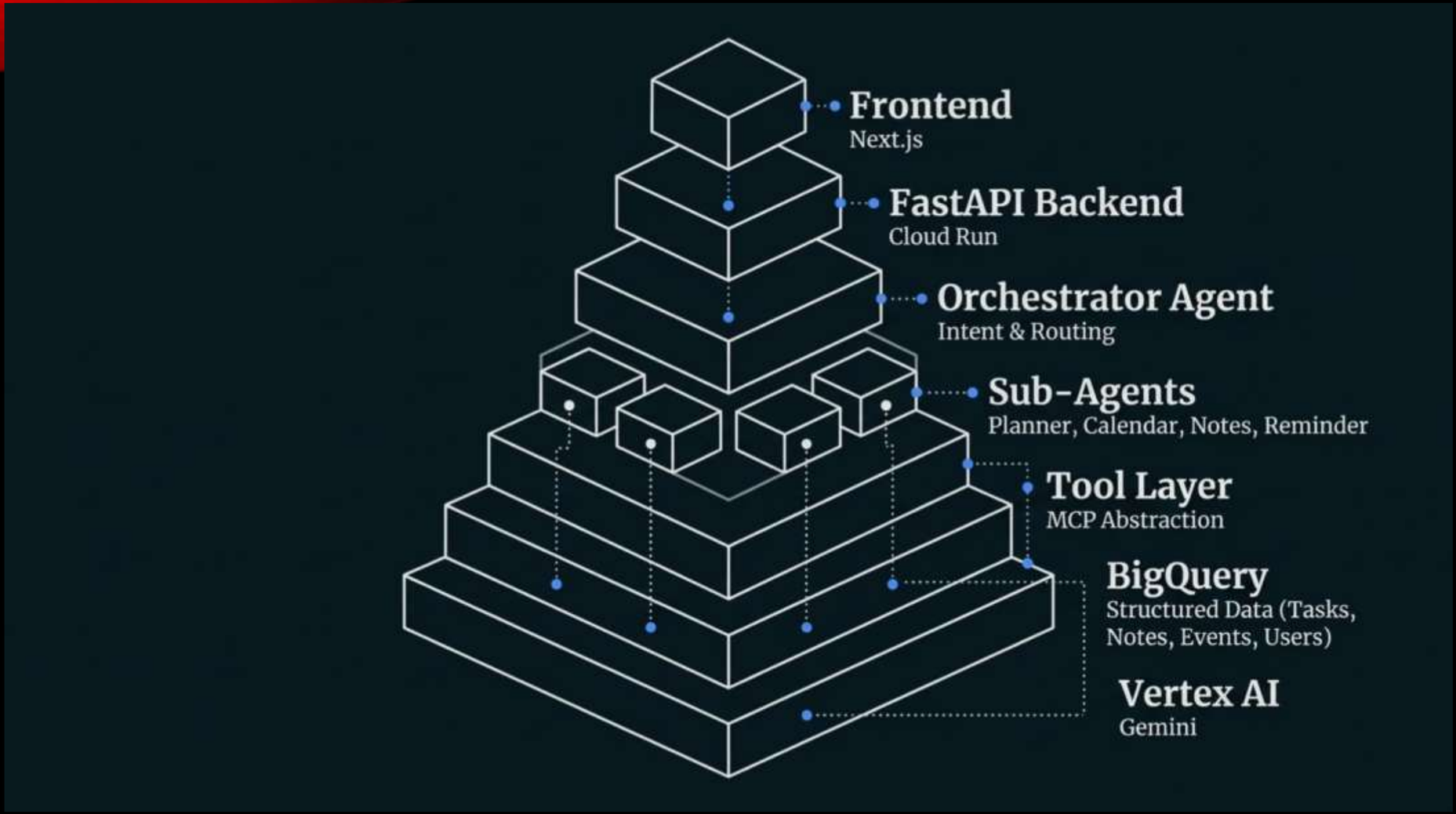
An interactive graph that renders the full agent decision path: Orchestrator → Agent → Tool → Output, making AI reasoning transparent.

UX Philosophy

Clean, minimal design with real-time feedback loops. Every interaction is instantly reflected in the dashboard, giving users full confidence that the system is acting on their behalf.

ARCHITECTURE

Modular, Scalable, Production-Ready



Modular Design

Each agent is independently deployable and swappable — enabling rapid iteration.

Async Processing

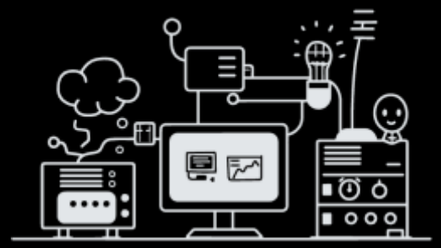
FastAPI's async backend ensures non-blocking, high-throughput agent coordination.

Extensible

New agents and tools can be plugged in without disrupting existing workflows.

TECHNOLOGY STACK

Powered by Google Cloud



Vertex AI (Gemini)

Drives all agent reasoning, intent classification, and decision-making. Gemini's advanced language understanding enables nuanced workflow orchestration across all agents.

BigQuery

Stores all structured outputs — tasks, notes, calendar events, and user profiles. Enables fast, scalable querying for workflow history and analytics.

Cloud Run

Hosts the FastAPI backend as a fully managed, auto-scaling containerized service. Delivers production-grade performance with zero infrastructure overhead.

Thank You — Google Cloud Gen AI Academy APAC 2026

> **"Transforming AI from a passive assistant into an active operations manager."**

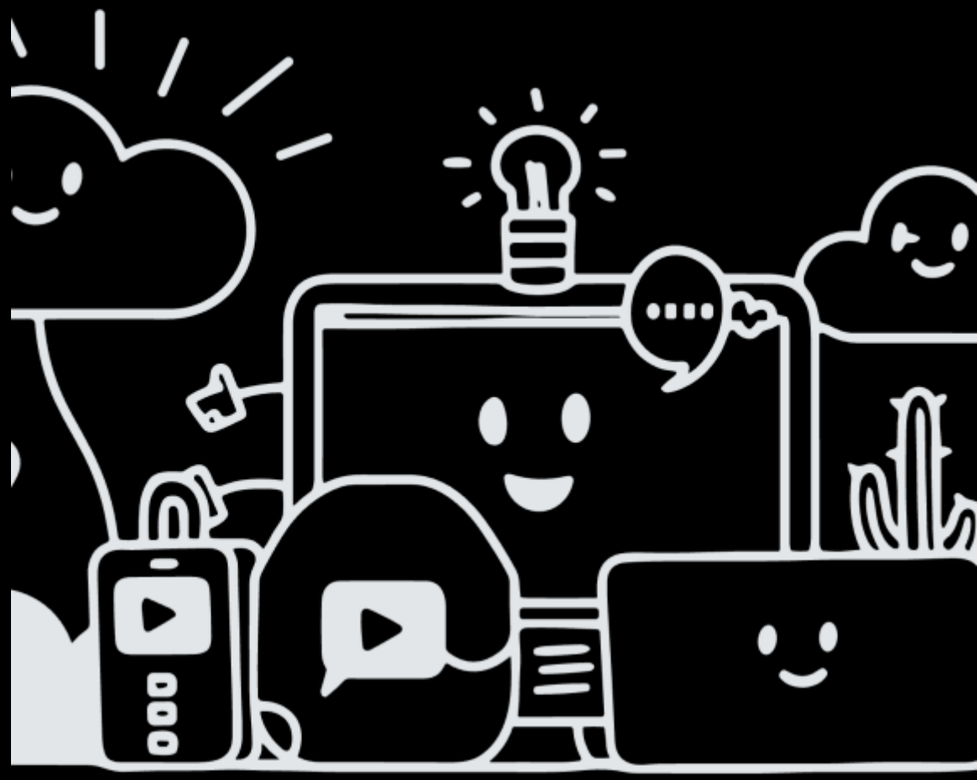
🔗 Project Links

GitHub Repository:
https://github.com/KetanDutt/GoogleCloudGenAIHackathon2026

Live Demo:
https://ai-ops-manager-frontend-dt4ppeftoa-uc.a.run.app

Built by

Ketan Dutt

Google Cloud Gen AI Academy APAC 2026 Multi-Agent AI Systems Track

VERTEX AI | BIGQUERY | CLOUD RUN | GEMINI